PERFORMANCE SCORE - 1 of 6

e-Office Score

File-disposal efficiency from the e-Office file-movement system - how much was cleared, how fast, how little was left pending, and how steadily it was done.

Clearance 25% - Low Pendency 10% - Disposal Speed 45% - Consistency 20%

A plain-language note - every formula matches the live calculation.

1 What this score measures and why it exists

Government runs on files. Every proposal, sanction, transfer, payment and clearance moves as an e-Office file that arrives in an officer's electronic in-tray, is examined, and is either decided or forwarded. The e-Office Score rates how efficiently that flow was handled. It rewards an office that cleared a LARGE workload, QUICKLY, leaving LITTLE pending, at a STEADY, predictable pace. The output is one number from 0 (poor) to 100 (excellent).

Crucially, the four quality indicators below are computed from the officer's OWN figures alone - they do not depend on who else happens to be on the screen or in the filter. This makes a score stable and defensible: an officer's measured clearance, pendency, speed and consistency never change merely because a colleague was added to or removed from a list. Only the final 'difficulty' comparison looks across officers.

The data taken from the e-Office system

Opening Balance (OB)	files carried over un-disposed at the start of the report period.
Files Received (FR)	new files that arrived during the period.
Files Processed (FP)	files disposed (decided or forwarded) during the period.
Files Pending	files still un-disposed at the end of the period.
Average time / file	the mean number of minutes taken per file.
Median time / file	the 'typical' (middle) minutes per file when all files are lined up.
Active days	the number of distinct days on which the officer actually worked files.

Two derived quantities are used throughout: the TOTAL WORKLOAD = OB + FR (everything the office had to deal with), and the DAILY PACE = total workload / active days (how intensely the office worked).

2 The four quality indicators (each 0-100)

The heart of the score is four indicators. Each turns a raw fact into a 0-100 sub-score, and each is finally clamped (capped) into the 0-100 range so that no single odd ratio can distort the blend.

Indicator 1 - Clearance Rate (weight 25%)

The single most natural question: of everything the office had to handle, what fraction did it actually dispose? A value of 90 means nine-tenths of the total workload was cleared.

INDICATOR 1

```
Clearance Rate = Files Processed / (Opening Balance + Files Received) x 100
```

Note the denominator is the TOTAL workload, not just receipts. Measuring against receipts alone would let an office that ignores its backlog look good; measuring against OB + FR closes that loophole.

Indicator 2 - Low Pendency (weight 10%)

The flip side of clearance: how small is the leftover pile relative to the whole workload? A value of 90 means only one-tenth of the workload remains pending at the end.

INDICATOR 2

```
Low Pendency = ( 1 - Files Pending / (Opening Balance + Files Received) ) x 100
```

Clearance and Low Pendency are related but not identical, because files can also be created or transferred during the period; keeping both, at 25% and 10%, captures both 'throughput' and 'residue' without double-counting.

Indicator 3 - Disposal Speed (weight 45%, the heaviest)

Timeliness is the core of e-Office performance, so speed carries the most weight. It is judged against a FIXED benchmark of 1200 minutes per file - meaning an office that truly averages 1200 minutes per file scores exactly 50 on this indicator; faster scores more, slower scores less. Using a fixed benchmark (rather than the cohort average) keeps the score absolute and stable. Two statistical safeguards make it fair:

- ROBUST TIME: time per file is taken as 0.65 x median + 0.35 x average. The median (the typical file) is trusted more than the average, because a handful of unusually large files can inflate the average and make a generally fast office look slow.
- SMALL-SAMPLE STEADYING (empirical-Bayes shrinkage): an office that processed only a few files has an unreliable speed - luck dominates. So K = 30 'pretend' files at the benchmark are mixed in. An office with thousands of files is barely affected; one with a handful is pulled towards the neutral benchmark until it has earned enough real evidence.

INDICATOR 3 - DISPOSAL SPEED

```
robust time tEff = 0.65 x median + 0.35 x average

                   n x tEff + K x REF      n = Files Processed
steadied tStar  =  --------------------     K = 30   REF = 1200
                       n + K

Disposal Speed = REF / (REF + tStar) x 100
               = 1200 / (1200 + tStar) x 100
```

The curve REF/(REF+tStar) is deliberately gentle: it rewards being fast without exploding to infinity for a near-zero time, and it cannot go below 0 or above 100. Edge handling: if only one of median / average is present, the missing one is set equal to the present one, so the formula always works.

Indicator 4 - Consistency (weight 20%)

A steady office is a well-managed office. Consistency rewards a predictable pace. When the average time sits close to the median, work is steady and the score is near 100. When the average is far ABOVE the median, a few very slow files dragged things out, and the score falls. The exponential curve maps any gap smoothly onto 0-100.

INDICATOR 4 - CONSISTENCY

```
gap = max( 0 , average - median ) / ( median + 1 )
Consistency = exp( - gap ) x 100

gap = 0   -> Consistency = 100  (perfectly steady)
gap = 0.7 -> Consistency ~ 50
```

The 'max(0, ...)' means an office whose average is BELOW its median (very even work) is treated as perfectly steady rather than rewarded beyond 100. The '+1' in the denominator avoids dividing by zero when the median is tiny.

3 Combining the four into the final score

Step A - Efficiency (the weighted blend)

The four indicators are blended using weights that add up to exactly 1.00. Disposal Speed dominates at 45%; Clearance adds 25%, Consistency 20%, and Low Pendency the remaining 10%.

STEP A - EFFICIENCY

```
Efficiency = 0.25 x Clearance
           + 0.10 x Low-Pendency
           + 0.45 x Disposal-Speed
           + 0.20 x Consistency
(result lies between 0 and 100)
```

Step B - the workload (difficulty) factor

Efficiency alone would let an office that perfectly cleared 50 files tie with one that perfectly cleared 5000. The workload factor corrects this. It combines the VOLUME of files (OB + FR) with the daily INTENSITY (files per active day), places both on a log scale against fixed full-credit targets (5000 files and 60 files per day), and converts the result into a factor whose floor is 0.50.

STEP B - WORKLOAD FACTOR

```
avail = OB + FR ;   pace = avail / active-days
load  = log(1 + avail) + 0.5 x log(1 + pace)
Lfull = log(1 + 5000) + 0.5 x log(1 + 60)
W     = clamp( load / Lfull , 0 , 1 )

Workload factor = 0.50 + 0.50 x W            (0.50 .. 1.00)
```

The lightest workload still keeps half credit (0.50), so a small but excellent office is dampened, never zeroed; the heaviest workload earns the full 1.00. The 0.5 weight on 'pace' lets daily intensity add to, but not dominate, raw volume. If 'active days' is missing, the factor falls back to pure volume.

The final score

FINAL e-OFFICE SCORE

```
e-Office Score = Efficiency x Workload factor          (0 - 100)
```

4 Worked examples

WORKED EXAMPLE - a strong, busy office

Opening 400, Received 4600 -> total workload = 5000 files
Processed 4500, Pending 500, median 900 min, average 1100 min, active 90 days

Clearance = 4500/5000 x100 = 90.0
Low Pendency = (1 - 500/5000) x100 = 90.0
tEff = 0.65x900 + 0.35x1100 = 970 min
tStar = (4500x970 + 30x1200)/(4500+30) = 971.5 min
Speed = 1200/(1200+971.5) x100 = 55.3
gap = (1100-900)/(900+1) = 0.222 ; Consistency = e^-0.222 x100 = 80.1
Efficiency = .25x90 + .10x90 + .45x55.3 + .20x80.1 = 72.4
avail 5000, pace 55.6 -> W ~ 0.98 ; factor = 0.50 + 0.50x0.98 = 0.99
e-Office Score = 72.4 x 0.99 = 71.7

WORKED EXAMPLE - a small office handled perfectly (why the floor matters)

Opening 20, Received 80 -> workload 100; Processed 100, Pending 0
median 600, average 620, active 20 days, n = 100
Clearance 100, Low Pendency 100
tEff=607, tStar=(100x607+30x1200)/130=743.7 ; Speed=1200/1943.7x100=61.7
gap=(620-600)/601=0.033 ; Consistency=96.7
Efficiency = .25x100+.10x100+.45x61.7+.20x96.7 = 82.1
Tiny workload -> W ~ 0.55 ; factor = 0.50 + 0.50x0.55 = 0.78
e-Office Score = 82.1 x 0.78 = 64.0
-> excellent quality, but the small load keeps it below the busy office.

5 The statistics, explained from scratch

Average versus median

The AVERAGE (mean) adds every value and divides by the count. The MEDIAN is the middle value once all are lined up smallest to largest. They differ when the data is lopsided: a few very slow files pull the average up but leave the median where it is. Because the median better represents the 'typical' file, Disposal Speed leans 65% on it and only 35% on the average - and Consistency literally measures how far apart the two are.

Empirical-Bayes shrinkage (steadying small samples)

Imagine an office that processed just 4 files, all quick by luck. Its raw speed looks spectacular but means little. Shrinkage blends the measured value with a neutral prior - here, K = 30 imaginary files sitting exactly at the 1200-minute benchmark. With thousands of real files the imaginary 30 are negligible; with only a few, they dominate and pull the office back to neutral. This is the standard statistical cure for unreliable small samples.

SHRINKAGE (GENERAL FORM)

```
steadied value = ( n x measured + K x benchmark ) / ( n + K )
```

Logarithms for size

A logarithm turns multiplication into addition: log(10) + log(10) = log(100). On a log scale, going ten-fold is always the same step. This is exactly what we want for 'size of workload' - an office handling 5000 files is meaningfully bigger than one handling 500, which is bigger than 50, in equal felt steps. A plain scale would make the 5000-file office dwarf everyone and flatten all the others to near zero.

Clamping (capping)

Every indicator and the factor input are forced into a sensible range (0-100, or 0-1). Real data occasionally produces impossible ratios (e.g. processing more than were received, due to timing). Clamping keeps the blend stable and the final number interpretable.

Weighting

The 25/10/45/20 weights encode policy: speed first, throughput next, steadiness, then residue. Because they sum to 1.00, the efficiency stays naturally on the 0-100 scale.

6 Reading it on the dashboard & FAQ

The e-Office page lists every officer with their four indicators, the Efficiency, the Workload factor and the final Score, and a state-wide Rank. A Normalised / Absolute switch lets you view the raw computed scores or the same scores stretched so the best officer reads 100.

Q. My clearance is 95% but my score is mid-range. Why?

A. Speed carries 45% of the weight and the difficulty factor scales the whole thing - high clearance alone cannot carry the score if files were slow or the workload was light.

Q. Does adding a filter change my score?

A. No. The four indicators are absolute (from your own figures). Only the normalised VIEW and the difficulty comparison look across officers, and the difficulty floor protects small offices.

Q. I processed very few files but very fast. Why am I not at the top of Speed?

A. Small-sample shrinkage pulls a tiny sample towards the benchmark, so a lucky few quick files cannot create a fake top speed.

Q. Why is the benchmark 1200 minutes?

A. It is a fixed reference at which Speed scores 50, chosen so the scale is stable over time. It is configurable by the administrator if policy changes.

G Glossary of terms

Opening Balance / Files Received
files carried over / newly arrived in the period.

Total workload
Opening Balance + Files Received - the denominator for throughput.

Clearance Rate
share of total workload disposed (Indicator 1).

Low Pendency
how little remains un-disposed vs the whole workload (Indicator 2).

Disposal Speed
pace per file vs the fixed 1200-min benchmark (Indicator 3).

Consistency
closeness of average to median - steadiness (Indicator 4).

Median
the middle value; robust to a few extreme files.

Empirical-Bayes shrinkage
pulling a small-sample figure towards a benchmark for reliability.

Efficiency
the weighted blend of the four indicators (Step A).

Workload factor
0.50-1.00 multiplier growing with files handled and daily pace (Step B).

Logarithm (log)
a scale on which equal multiples are equal steps; used for size.

Clamp / cap
forcing a value into a fixed range (here 0-100 or 0-1).

All weights, benchmarks and difficulty-factor floors/ceilings are stored in the score_config table and are configurable by the administrator in the Score-Weights page of the dashboard, so the system can be re-tuned without changing any code. Scores recompute automatically whenever the underlying data changes. Every formula in this note matches the live calculation exactly.